MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) December 1st, 2012
Instructions: Attempt <u>any three out of the following four questions</u> on both pages. <u>Mention which three you would like to have graded</u>. Open book and open notes.

1. An investor must choose between buying two buckets each with 25 rocks of gold. He randomly chooses 5 rocks from Bucket A and 5 rocks from Bucket B and calculates the grams of gold in each of the rocks with the following results

   BUCKET A - 81, 75, 92, 108, 77
   BUCKET B - 59, 64, 81, 83, 63,

   Use a large sample approximation for your answers below and state your assumptions

A. Using only the data from Bucket A, Give a point estimate and 95% confidence interval for the total amount of gold in bucket A
B. The investor believes that there is <u>at least 175</u> <u>more units of gold in bucket A than Bucket B.</u> Statistically test this hypothesis. at $p = 0.01$

2. Researchers seek to estimate changes in the total number of polar bears in particular regions. Polar bears hunt on floes of ice in the seas off the coast of the "Artic ring of life" where seals are abundant. One region is by the coast of the Southern Beaufort Sea. This is an area, 1000 kilometers in length by 200 kilometers in width (land on coast plus adjacent sea). The 200,000 square kilometer region was divided into a grid of 8000 subsquares, each 5 kilometers in length by 5 kilometers in width. In 1986 a random sample of 400 subsquares were flown over, and a total of 90 polar bears were observed. In 2006 an independent random sample of 400 subsquares were flown over, and a total of 75 polar bears were observed. You can assume that the polar bears are distributed completely at random over the region.

a) Estimate the total number of polar bears in the region in 1986, and in 2006.

b) For each estimate, give a standard error of the estimate
c) Test the null hypothesis at the 0.10 level of significance of no change (between 1986 and 2006) in number of polar bears in the region, versus the research hypothesis that there has been a decrease in the number of polar bears.

3. Scientists from Rutgers have built a spaceship to go to Saturn, but forgot that the space ship had to go through the asteroid belt. The spaceship can survive a single impact from an asteroid of at most 1,000 Newtons.

   - <u>Assume the Newton impact from each asteroid hitting the spaceship is distributed normally with u = 750 and σ = 200.when answering the questions below</u>

- If the ship is not destroyed by an asteroid, it recovers completely. There is no residual or additive damage. A hit from an asteroid that does not destroy a ship does not weaken the ship for the next asteroid hit
- Each asteroid hit is independent
- Only hits by asteroids prevent the ship from surviving.

A. Assume the ship is destined to be hit by 10 asteroids on a trip as long as it survives. What is the probability it survives this trip?
B. How likely is the ship to be destroyed on the impact of a 5th asteroid? You can assume that the ship will be hit by at least 5 asteroids (if it survives that long)
C. How many Newtons would the ship need to be able to survive from a single hit to have an 80% chance to survive a trip with exactly 10 impacts?

4. A researcher wants to compare different treatments to reduce cholesterol levels. The response variable is the reduction in cholesterol level. (You can assume normality and homogeneity of variance.). The treatments include different combinations of the factors: Diet and Exercise.

Diet has 2 levels: Low Sugar/carb Diet, High fruit/vegetable Diet.

Exercise has 3 levels: Walking, Jogging, Zumba dancing.

Eight cardiologists participate in the study, and are divided at random into two groups of 4 cardiologists each. For each cardiologist there are 18 patients. (a total of 144 patients). All the patients of a given cardiologist have the same diet. For 4 of the cardiologists the patients' diet was Low Sugar/Carb, for 4 of the cardiologists the patients' diet was High fruit/vegetable. For each cardiologist the 18 patients were divided at random into three groups of 6 patients each. 6 patients assigned to Walking, 6 assigned to Jogging, and 6 assigned to Zumba dancing.

Show the ANOVA Table, showing Source of Variability, degrees of freedom,

Expected Mean Squares, and show which F Tests to take.